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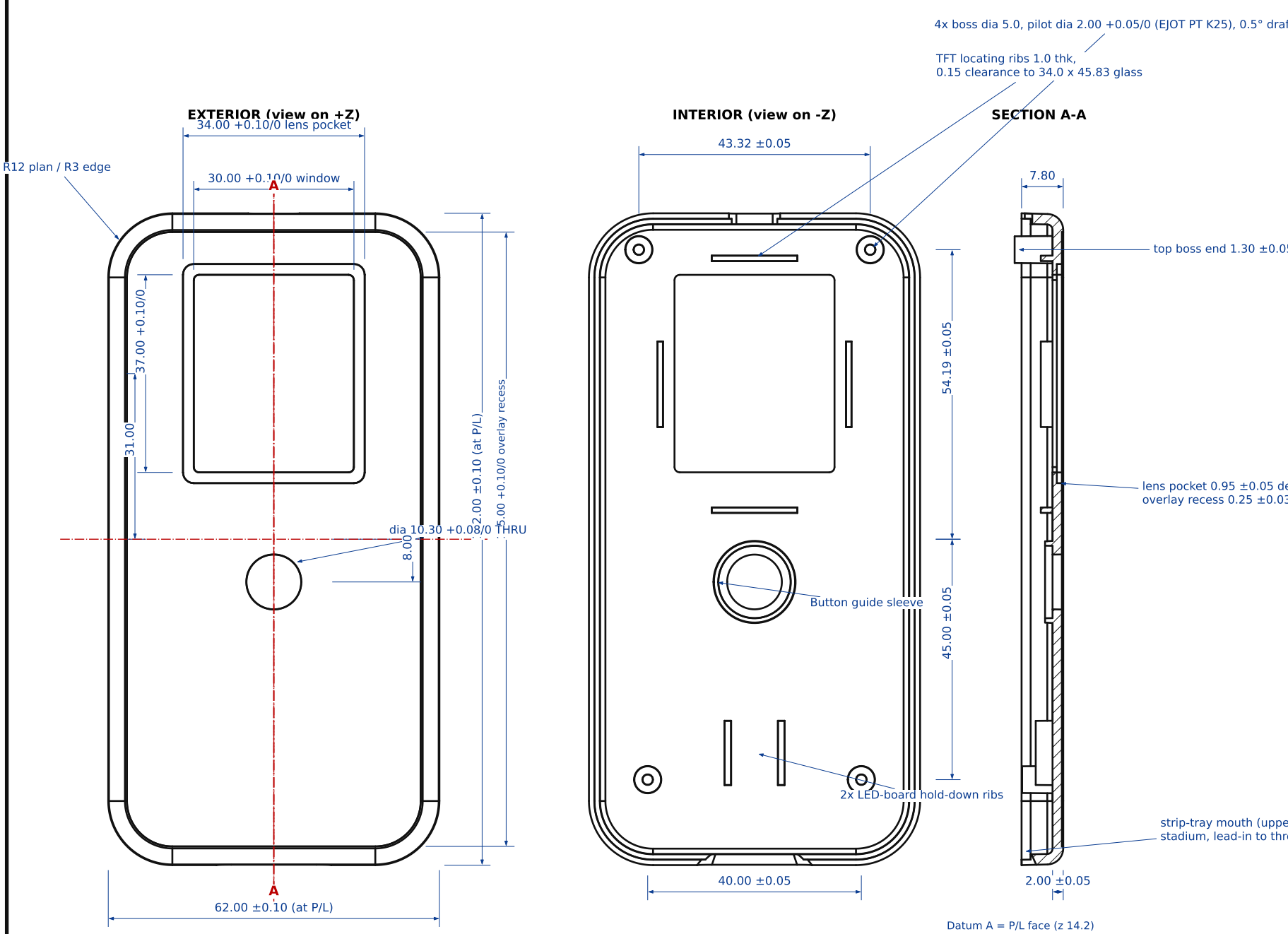
Tray handle = light-seal plug

NOTES

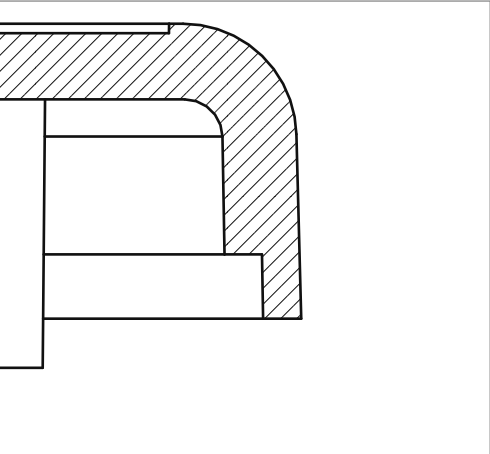
1. Datum A = parting plane (P/L), z = 14.2 from the rear skin. B = centre line X. C = centre line Y.
2. All plastics: ABS, should shrink 0.5 % built into the tool STEP data (see HS-300 sheets).
3. Parting plane runs through the USB-C centre and the strip-tray mouth: No slides / lifters in any tool.
- 3a. SINGLE BUTTON. Strip goes in a slide-in tray (HS-106): load strip outside, push tray home (hard stop + detent click); the tray end stop and the read aperture are on the same black part (HS-105).
4. Housing closure: rim lap joint + 4 x PT K25x16 from the rear. Top two screws pass through the PCB top holes (PCB NOT clamped: 0.30 hold-down gap). Bottom PCB holes: 2 x PT K25x6.
5. White ABS is translucent: the optical path is enclosed in the BLACK optical block (HS-105).
6. Tolerances: general per sheet 6; critical fits and 15 stack-ups on sheet 6.
7. Colour variants (teal / grey / purple ...) come from the printed overlay only - one set of tools.
8. OPEN ITEMS: confirm PCB component heights (<= 3.0 below, TFT FPC route), USB-C overhang 3.0 mm past PCB edge, tact switch at (0, -8), HB STRIP SIZE (6.0 x 30 x 0.5 assumed), LiPo 503035 wiring.

LEV RAM LIFESCIENCES		HemoSure Hb meter casing
GENERAL ARRANGEMENT, STACK-UP & BILL OF MATERIALS		Drawn: Claude (AI) for S. Saboo Chk: ____ Appr: ____
PART HemoSure casing assembly	DWG NO. HS-100	REV A
MATERIAL see BOM	SCALE 1:1 / NTS	SHEET 1 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS 42 g (plastics)	DATE 2026-09-28	STATUS DRAFT - DFM review

A
B
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DETAIL B - LAP STEP (5:1, section y = 0)



CRITICAL DIMENSIONS (HS-101)

Feature	Nominal	Tol.	Steel-safe
Outer profile at P/L	62.00 x 122.00	+0.10 / -0.10	-
Display window (at pocket floor)	30.00 x 37.00	+0.10 / 0	cut small
Lens pocket	34.00 x 41.00	+0.10 / 0	cut small
Lens pocket depth	0.95	+0.05 / -0.05	cut shallow
Overlay recess	55.00 x 115.00	+0.10 / 0	cut small
Overlay recess depth	0.25	+0.03 / -0.03	cut shallow
Button hole	dia 10.30	+0.08 / 0	cut small
Screw pilot (PT K25)	dia 2.00	+0.05 / 0	pin small
Top boss end below P/L	1.30	+0.05 / -0.05	leave long
Bottom boss end above P/L	0.10	+0.05 / -0.05	leave long
Lap step width x depth	1.00 x 1.70	+0.05 / 0	cut narrow
Boss pattern (both halves)	43.32 x 70.13	+0.05 / -0.05	-

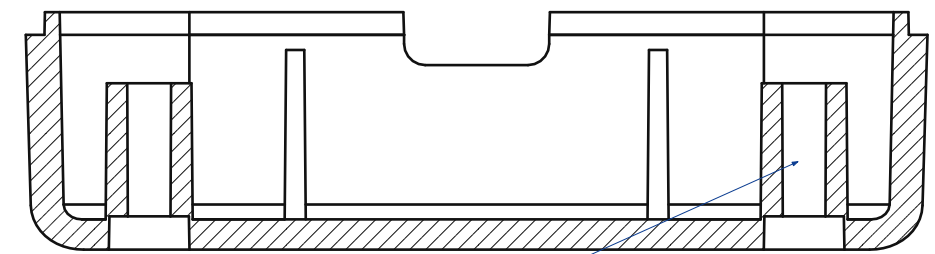
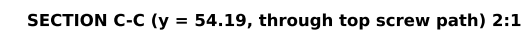
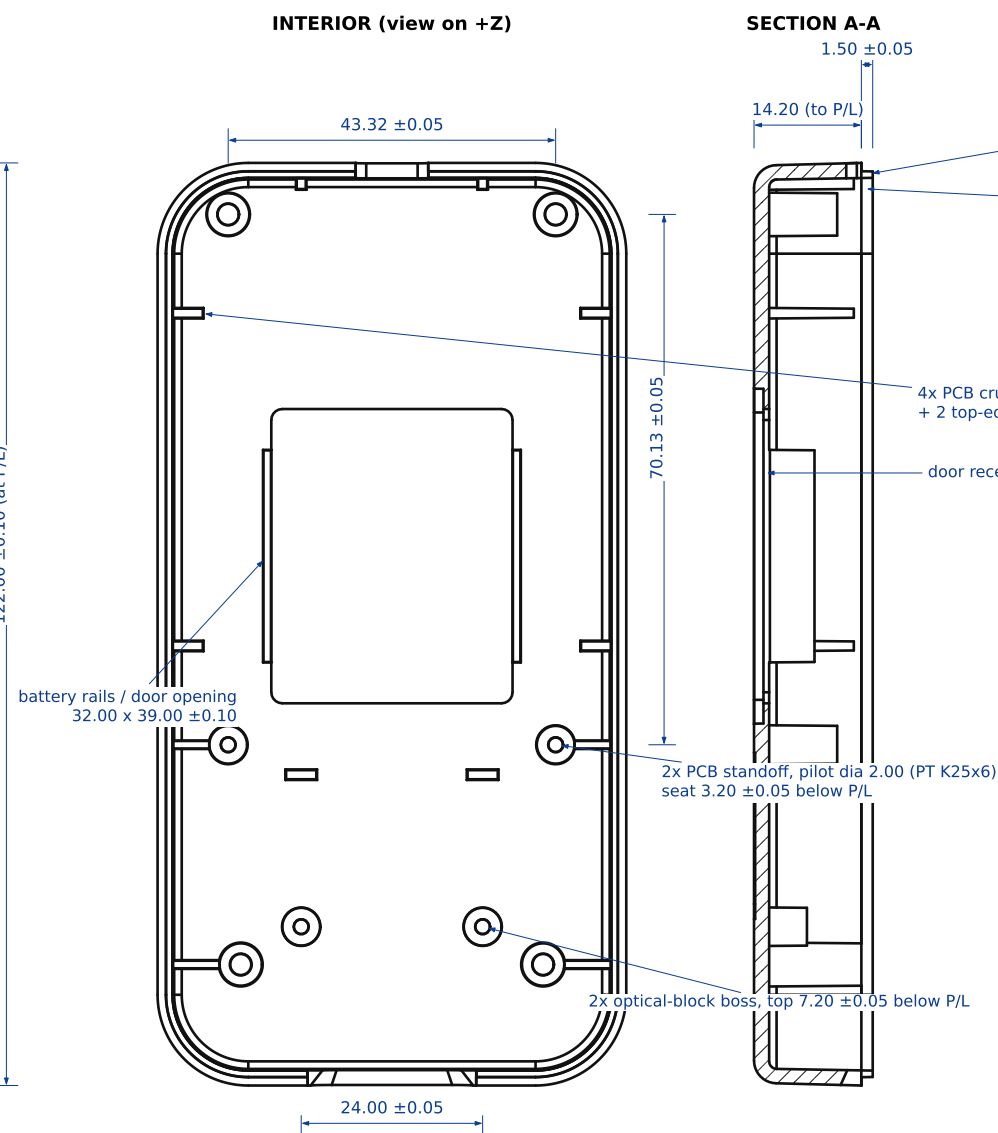
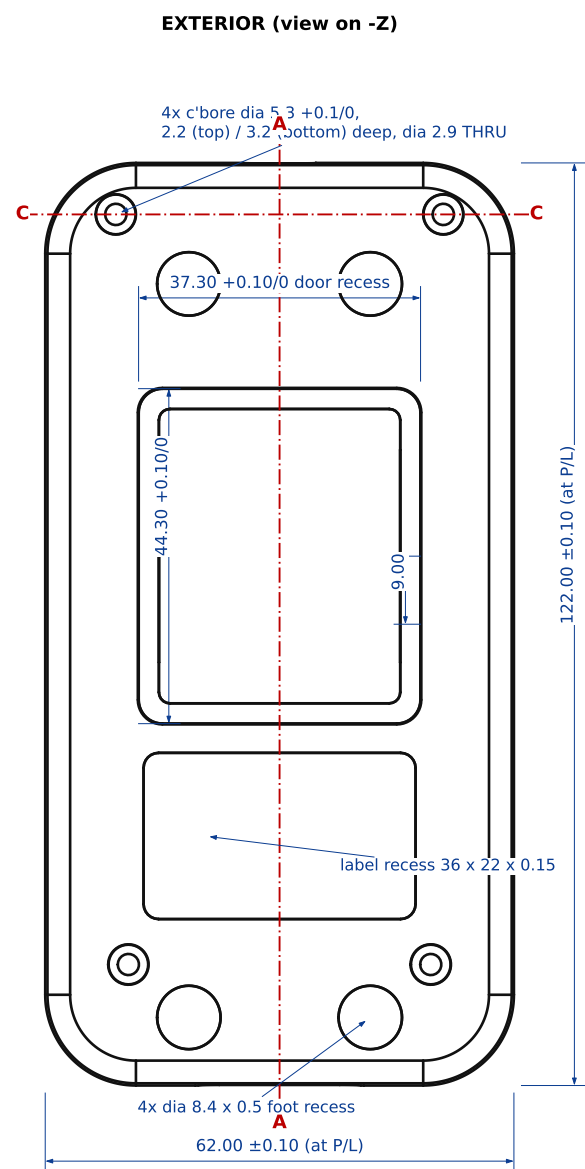
NOTES

- Material ABS natural white (colour per overlay). Mould shrink 0.5 % (tool data). Wall 2.00 nominal.
- Draft: outer walls 1.5° (texture VDI 24), inner walls 1.0°, bosses/ribs 0.5°/side - modelled, do not add.
- Ribs 1.0-1.2 thk (<= 60 % wall) - no sink on the cosmetic face. Boss OD 5.0 / pilot 2.0 (wall 1.5).
- P/L flatness 0.10. Outer profile at P/L to match HS-102 within 0.10 per side (step).
- No flash on P/L, lap step and window; no sink / flow marks / weld line on the overlay area.
- Gate: 1 x sub (tunnel) gate dia 1.2 into inner +X wall at y = 0 (hidden). Ejector marks core side only, flush +0/-0.05.
- General tolerances: sheet 6 table (ISO 20457-style). Fits marked here override.

LEVRAM LIFESCIENCES
FRONT HOUSING

HemoSure Hb meter casing
Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART Front housing	DWG NO. HS-101	REV A
MATERIAL ABS natural white (LG HI-121H / PA-757 eq.)	SCALE 1:1	SHEET 2 / 11
FINISH Out VDI 3400 Ref 24 / In SPI B-2	UNITS mm	PROJ. 1st angle (ISO E)
MASS 15.0 g	DATE 2026-09-28	STATUS DRAFT - DFM review



screw path: c'bore -> standoff
(seat z 11.0) -> PCB -> front boss

CRITICAL DIMENSIONS (HS-102)

Feature	Nominal	Tol.	Steel-safe
Boss pattern (both halves)	43.32 x 70.13	+0.05 / -0.05	-
Outer profile at P/L	62.00 x 122.00	+0.10 / -0.10	-
PCB seat below P/L	3.20	+0.05 / -0.05	leave long
Tongue thickness x height	0.90 x 1.50	0 / +0.05 / -0.05 / -0.05	cut thick
USB-C opening (split on P/L)	9.60 x 4.00	+0.10 / 0	cut small
Door recess	37.30 x 44.30	+0.10 / 0	cut small
Door recess depth	1.25	+0.05 / -0.05	cut shallow
Screw clearance / c'bore	dia 2.9 / dia 5.3	+0.10 / 0	-
PCB crush-rib tip to PCB edge	0.05	+0.05 / -0.05	cut tight
Optical-block boss top below P/L	7.20	+0.05 / -0.05	leave long

NOTES

1. Material ABS natural white, shrink 0.5 %. Wall 2.00; floor 2.00; door ledge 0.75 (reinforced by rails).
2. Draft as HS-101 (modelled). Tongue 1° both faces. Recesses 1°. Pins/holes 0.25-1°.
3. PCB located by crush ribs (+/-0.05) and 2 PT K25x6 at the bottom holes; top holes carry the housing screws.
4. USB-C and strip-tray mouth split on the P/L: half in each housing, no side action.
5. Gate: 1 x sub gate dia 1.2 into inner +X wall at y = +20 (hidden). Ejection: sleeves on all bosses.
6. No sink opposite bosses / rails on the rear skin (bosses covered, gussets 1.0 thk).
7. Label recess and foot recesses: keep texture-free (SPI B-2) for adhesion.

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

REAR HOUSING

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART	DWG NO.	REV
Rear housing	HS-102	A
MATERIAL	SCALE	SHEET
ABS natural white (LG HI-121H / PA-757 eq.)	1:1	3 / 11
FINISH	UNITS	PROJ.
Out VDI 3400 Ref 24 / In SPI B-2	mm	1st angle (ISO E)
MASS	DATE	STATUS
22.3 g	2026-09-28	DRAFT - DFM review

A



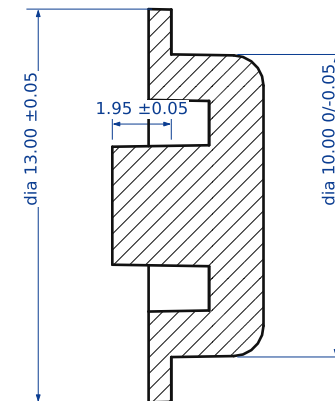
C



HS-103 SECTION (x = 0)



HS-104 BUTTON - section 4:1



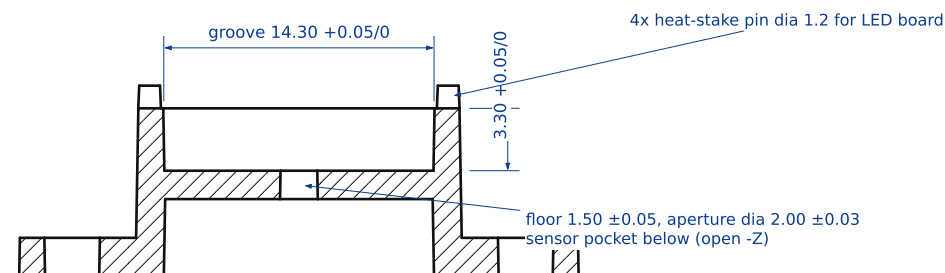
NOTES

HS-103 door: ABS white, same resin lot as HS-102.
 Flush 0/0.15 to rear skin; gap 0.15-0.25/side.
 HS-104 button: ABS black (or TPU 90A if softer
 feel wanted - same lot). Pad-print power icon.
 HS-105 optical black: ABS BLACK, opaque at 2 mm
 (transmission < 0.1 % @ 400-700 nm - verify).
 Groove Ra <= 0.8 (SPI B-1) to avoid scratching
 the strip tray. Parting at underside.
 Add 0.5 x 45° lead-in at groove entry (T1).
 All: draft 1° modelled on every wall.

Cap rests on the tact actuator; 0.45 play to the housing face (stack 3).

Mould windows under hooks/bump let the cavity form the hook undersides (no lifters).

HS-105 SECTION D-D (y = -40, optical axis) 2.5:1

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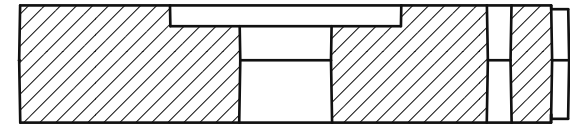
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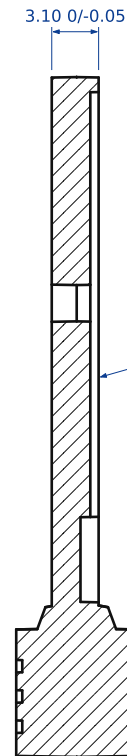
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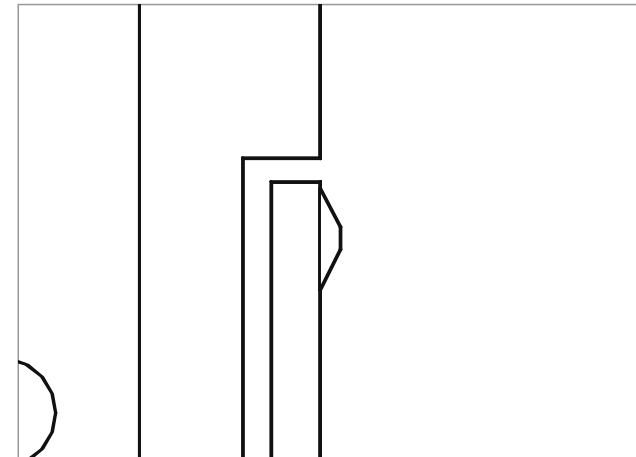
SECTION D-D at the read axis (y = -40) 5:1



SECTION A-A (x = 0)



DETAIL E - DETENT ARM + BUMP (6:1)



NOTES

HOW IT WORKS (single-button meter)

1. Pull the tray out by the handle.
2. Drop the strip into the pocket, read window down, pushed against the front wall (pocket locates it to +/-0.05).
3. Slide the tray in: it rides on the black optical block, hits the end stop and CLICKS (detent bump into notch).
The detent sits 0.15 up the ramp -> it pushes the tray onto the stop, so the strip position is set by the stop.
4. The handle plug closes the mouth (light seal).
5. Press the one button -> measure. Tray lift ≤ 0.4 , lateral ≤ 0.15 , axial window error ≤ 0.3 (stacks 15-22).

STRIP SIZE IS ASSUMED 6.0 x 30 x 0.5 - one parameter each.

MOULDING

ABS black, 1° draft both ways from the P/L (z 14.2), parting through the handle centre like the housings.
Detent arm 1.1 x 13 mm, bump 0.45 / notch 0.40: strain 0.4 % (ABS ~2.5 %) -> many thousand cycles.
Core-outs under the pocket walls keep walls ~1.1 (no sink on the sliding faces).

LEVRAM LIFESCIENCES

STRIP TRAY (SLIDE-IN, SINGLE-POSITION)

HemoSure Hb meter casing

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART Strip tray	DWG NO. HS-106	REV A
MATERIAL ABS black (2 % carbon MB)	SCALE 2:1	SHEET 5 / 11
FINISH SPI B-1 sliding faces	UNITS mm	PROJ. 1st angle (ISO E)
MASS 3.1 g	DATE 2026-09-28	STATUS DRAFT - DFM review

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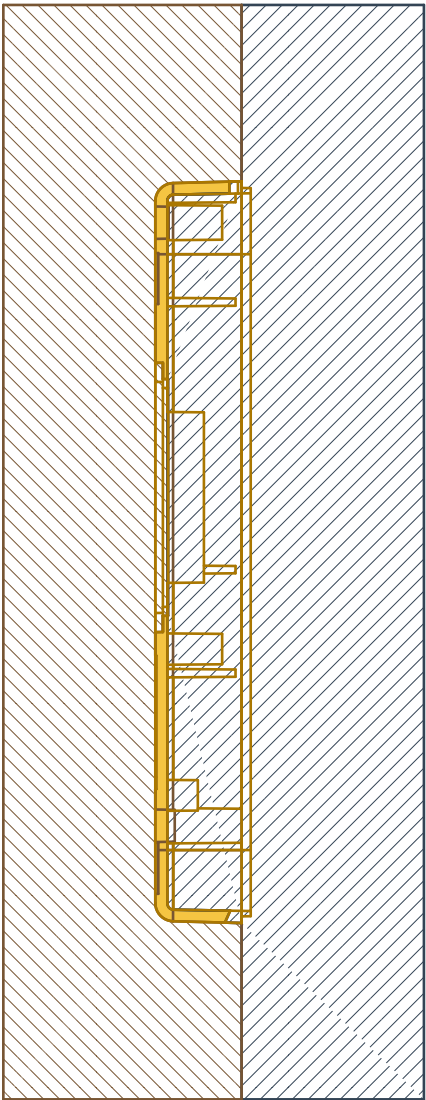
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REAR HOUSING - core (blue /) + cavity (brown \), part yellow

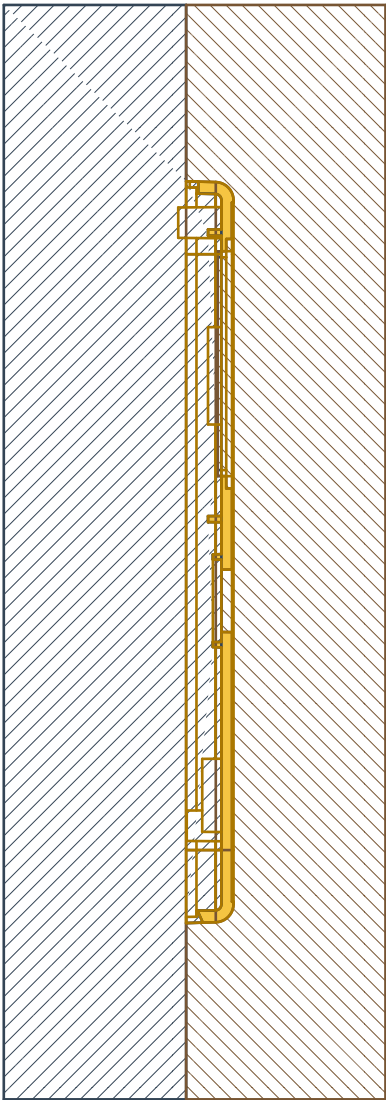
Section x = 0 at 0.8:1; mould opens along Z (horizontal on this sheet)



core bbox [120.6, 180.9, 42.4] cavity bbox [120.6, 180.9, 39.4] (mm)
rays 98882 | undercut rays 0 | out-of-order steel 0

FRONT HOUSING - core (blue /) + cavity (brown \), part yellow

Section x = 0 at 0.8:1; mould opens along Z (horizontal on this sheet)



core bbox [120.6, 180.9, 36.0] cavity bbox [120.6, 180.9, 33.0] (mm)
rays 100140 | undercut rays 0 | out-of-order steel 0

NOTES

1. Inserts are the full block less the part (x 1.005). Toolmaker adds: pocket fits (H7/g6 to the plates), locks / taper interlocks (4 x 5° side locks per insert), ejector & sleeve holes, cooling drillings, vents, sprue/runner/gates.
2. Shut-offs: USB-C opening and cuvette mouth are formed by core steel shutting off on the P/L - give them 1-3° in the tool if the T1 shows scuffing (geometry change <= 0.03, inside the opening tolerance).
3. Leave steel-safe stock on: pilot pins -0.03, button hole -0.05, window/lens/overlay pockets -0.05, boss ends +0.05.
4. Polish core SPI B-2; cavity EDM texture VDI 24 after dimensional sign-off; lens pocket floor SPI A-3.

LEVRAM LIFESCIENCES

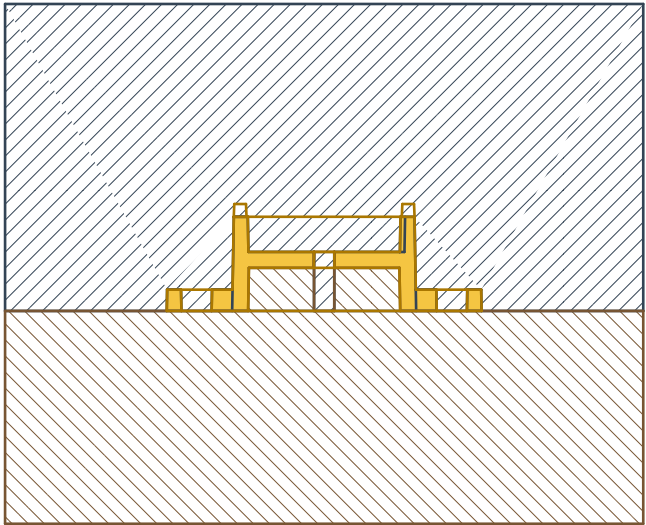
HemoSure Hb meter casing

MOULD TOOL A: CORE / CAVITY SECTIONS

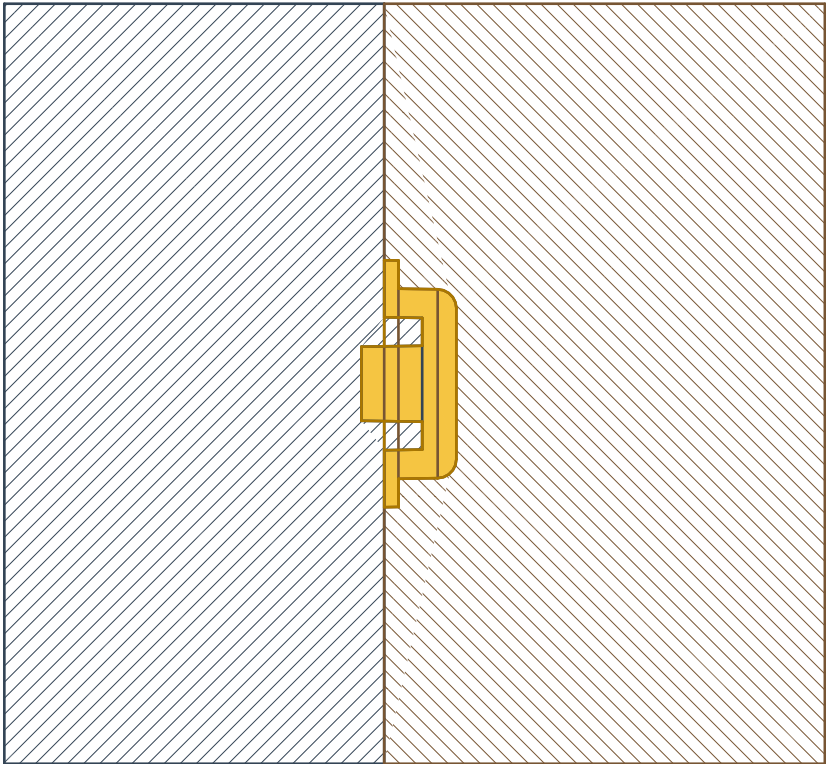
Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART Tool A inserts	DWG NO. HS-302	REV A
MATERIAL 1.2311 / 1.2738	SCALE 1:1	SHEET 9 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review

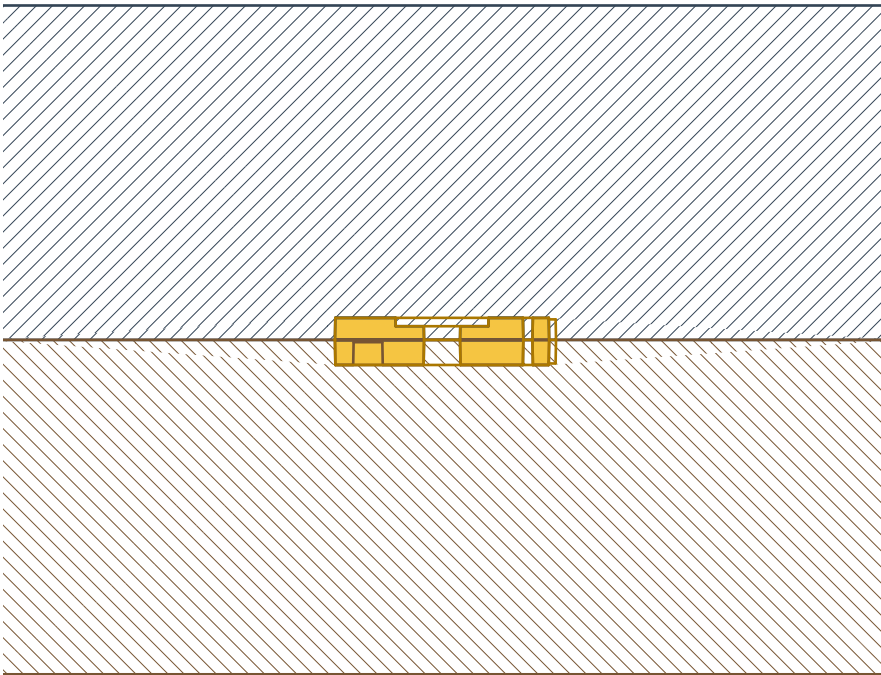
OPTICAL BLOCK - core / cavity (1.4:1)



BUTTON CAP - core / cavity (2.5:1)



STRIP TRAY - core / cavity (2.0:1)



rays 10809 | undercut 0 | out-of-order 0

rays 2117 | undercut 0 | out-of-order 0

NOTES

- All three parts black -> one family tool; if the button must be another colour, split it to its own 1-cavity tool.
- 1a. Tray + block are a matched pair: T1 fit check = tray slides with <= 2 N, clicks home, no rattle > 0.15 mm.
2. Optical block groove is the optical-path datum: hold 14.30 +0.05/0 and floor 1.50 ±0.05.
3. Heat-stake pins dia 1.2: vent the pin tips (0.01) to avoid short shots.
4. Verify opacity on T1 parts: no visible light through 2 mm with a torch in a dark room; spectrometer < 0.1 %.

Item	Tool B data
Cavities	button + optical block + strip tray, black ABS
rays 11573 undercut 0 out-of-order 0	
Parting	Button: flange u/s. Block: u/s. Tray: P/L z 14.2
Shrink	0.50 % (in insert STEP)
Feed	Sub gate dia 0.8 (button, on stem side), sub gate dia 1.0 (block, on ear underside)
Ejection	Button: sleeve on stem; block: 4 pins on ears + 2 pins on the end walls
Polish	Groove, tray faces SPI B-1 (sliding)
Proj. area	~7.5 cm² -> 30-50 t press
Feed (tray)	tray: sub gate dia 1.0 on the handle underside
Shot	5.6 g parts + ~3 g runner

LEVRAM LIFESCIENCES

HemoSure Hb meter casing

MOULD TOOL B (1+1+1, BLACK ABS)

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART	DWG NO.	REV
Tool B: button + optical block + tray	HS-303	A
MATERIAL	SCALE	SHEET
Inserts 1.2311 P20 / cavity 1.2083 for polish	2:1	10 / 11
FINISH	UNITS	PROJ.
-	mm	1st angle (ISO E)
MASS	DATE	STATUS
-	2026-09-28	DRAFT - DFM review

AUTOMATED CAD CHECKS (this revision)

Part	Vol. cm³	Mass g	Rays cast	Undercut rays	Steel out-of-order	Faces < 0.2° draft (>1 mm)	Parting z
Rear Housing	21.25	22.3	98882	0	0	0	14.2
Front Housing	14.28	15.0	100140	0	0	0	14.2
Battery Door	1.97	2.1	26150	0	0	0	1.2
Button Cap	0.29	0.3	2117	0	0	0	18.8
Optical Block	2.01	2.1	10809	0	0	0	7.0
Strip Tray	2.98	3.1	11573	0	0	0	14.2

Assembly interference check, all part pairs incl. PCB/TFT/USB-C/tact/LiPo/strip/tray/LED board: [['optical', 'tray', 0.022]] (optical x tray = designed 0.05 detent preload)

METHOD

How the checks work (hemasure_casing.py + mold_tools.py, re-run on every change):

- Undercut: vertical rays on a 0.25 mm grid; any ray crossing the part in 2+ intervals = trapped steel (validated on a test part).
- Inserts: rays through core, cavity and part; order bottom->top must be one steel half, part, other half (no interlock).
- Draft: every face's angle to the draw axis; faces taller than 1 mm with $< 0.2^\circ$ are listed (target: none).
- Stack-ups: 22 worst-case + RSS stacks, sheet 7. All pass worst case at this revision.

DESIGN RULES APPLIED

Wall 2.00 nominal (ABS range 1.2-3.5); thinnest: lens-pocket ledge 0.80, door ledge 0.75, lap skirt 1.00.

Ribs 1.0-1.2 (50-60 % of wall), rib height $\leq 3 \times$ wall except PCB posts (gusseted, sleeve-ejected).

Boss OD 5.0-5.6 / hole 2.0-2.9: boss wall 1.35-1.5 ($\leq 0.75 \times \text{wall}$) -> no sink on the rear skin.

All inside corners $R \geq 0.5$ at wall/floor (R1.0 modelled at the main wall-floor junction).

Flow length / wall (L/t) max ~42 from a single side gate - well inside ABS limit (~150).

Critical cosmetic: front overlay field, rear skin, P/L line. Non-cosmetic: interior, optical block.

MOULDING START POINT

Process window (start point): melt 230 °C, mould 55 °C, injection speed medium, hold 60-70 % of fill pressure for 4-5 s.

cooling 9 s, ejection at part temp < 85 °C. Gate freeze study to set the hold time. Regrind <= 15 % on cosmetic parts.

DATA PACKAGE

Files (all regenerated from `cad/hemosure_casing.py` - change a parameter, re-run, everything updates):

cad/HemoSure_*.step (parts + assembly) mold/HemoSure_*_{CORE,CAVITY}.step (steel size) drawings/*.pdf

cad/tolerance_stack.py (stacks) cad/dfm_checks.py (draft) mold/mold_check_report.json (ray undercut / insert order)

VALIDATION

T0/T1 TRIAL CHECKLIST (for the tool room)

1. Dry ABS 80 °C / 3 h; dew point <= -20 °C.
2. Short-shot study -> fill balance of the family tool.
3. Measure the critical-dims table (sheet 6) on 5 shots/cavity;
Cpk >= 1.33 on: boss pattern, PCB seat, window, lens pocket.
4. Assemble with a real PCB + TFT + LiPo: USB-C plug insertion,
button click (no pre-pressure, 0.25 travel), door open/close x 50.
5. Drop test 1.0 m x 6 faces (IEC 61010-1 / 60068-2-31 style).
6. Light-leak test of the optical cell with the strip tray inserted.
7. Disinfectant wipe x 500 (70 % IPA, 0.5 % NaOCl): no ESC cracks
at bosses (if cracking: switch to PC/ABS, shrink 0.5-0.7 %).
8. Then texture the cavities (VDI 24) and re-check P/L step.

LEV RAM LIFESCIENCES

HemoSure Hb meter casing

DFM VERIFICATION & MOULDING SPECIFICATION

Drawn: Claude (AI) for S. Saboo | Chk: ____ | Appr: ____

PART All moulded parts	DWG NO. HS-310	REV A
MATERIAL ABS	SCALE -	SHEET 11 / 11
FINISH -	UNITS mm	PROJ. 1st angle (ISO E)
MASS -	DATE 2026-09-28	STATUS DRAFT - DFM review